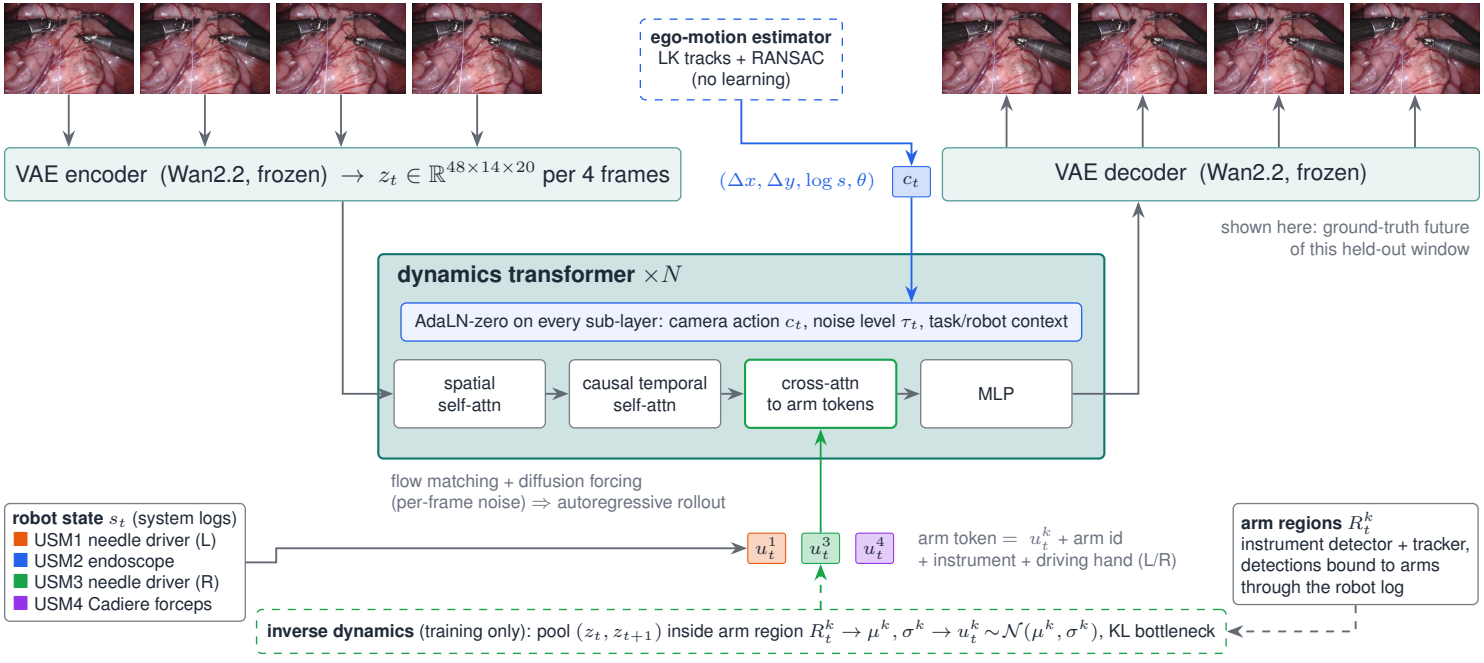

input: context frames $x_{1:4}$ (0.4 s apart) + actions
output: predicted frames $\hat{x}_{5:8}$ (next 1.6 s)
ego-motion estimator
LK tracks + RANSAC
(no learning)
VAE encoder (Wan2.2, frozen) → $z_t \in \mathbb{R}^{48\times14\times20}$ per 4 frames
$(\Delta x, \Delta y, \log s, \theta)$
c_t
VAE decoder (Wan2.2, frozen)
shown here: ground-truth future
of this held-out window
dynamics transformer $\times N$
AdaLN-zero on every sub-layer: camera action c_t, noise level τ_t, task/robot context
spatial
self-attn
causal temporal
self-attn
cross-attn
to arm tokens
MLP
flow matching + diffusion forcing
(per-frame noise) ⇒ autoregressive rollout
robot state s_t (system logs)
USM1 needle driver (L)
USM2 endoscope
USM3 needle driver (R)
USM4 Cadiere forceps
u_t^1
u_t^3
u_t^4
arm token = u_t^k + arm id
+ instrument + driving hand (L/R)
arm regions R_t^k
instrument detector + tracker,
detections bound to arms
through the robot log
inverse dynamics (training only): pool (z_t, z_{t+1}) inside arm region $R_t^k \to \mu^k, \sigma^k \to u_t^k \sim \mathcal{N}(\mu^k, \sigma^k)$, KL bottleneck